

| Question 6 |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
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| 6.         |  |  |  | <p>Indicative content of this level is...</p> <p><b>Process of geographical isolation</b><br/>Description of Geographical isolation e.g. when a population is separated by a physical <u>barrier</u><br/>Ice as isolating mechanism or description of which {prevents/ interferes with} gene flow/ demes {Founder effect/ or description of} on gene pool/genetic drift/ref to random mutation<br/>Effect of selection pressure changing the frequency of alleles in a population<br/>Eg of selection pressure for bears eg food availability/hunting new prey/camouflage/ability to metabolise a high fat diet</p> <p><b>Polar bear has adapted to live in its habitat.</b><br/>Selection of {alleles/ genes/ ref to change in genetic material} allowing a high fat diet<br/>The physiological changes had to be {quick/ref to less than 20,500 generations} if they were to subsist on a dangerously high fat diet<br/>Bears have evolved to eat (primarily) blubber /Nursing cubs rely on high fat milk<br/>Polar Bears must have large fat deposits under their skin to keep warm in Arctic conditions/Rely on metabolic water, a by-product of the breakdown of fat<br/>(Change in) fur colour (brown to white) so able to {hunt/camouflage}/The development of a sleeker body for hunting/swimming</p> | 4               | 5   |     | 9     |       |      |

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|            |  |  |  | <p><b>Two different species</b><br/>Not separate species as can interbreed to produce fertile offspring<br/>Gene pool still sufficiently similar<br/>Subsequent interbreeding of hybrids with Brown Bears<br/>Overlapping territories (due to receding ice) allows the 2 groups contact for interbreeding<br/>Will continue to interbreed so gradual loss of pure Polar Bears<br/>May outcompete pure Brown Bears/Loss of genetic biodiversity</p> <p><b>7-9 marks</b><br/>Detailed coverage of all three sections<br/><i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i></p> <p><b>4-6 marks</b><br/>Some detail of two sections or less coverage of three sections<br/><i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i></p> |                 |     |     |       |       |      |

| Question 6 |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks available |     |     |       |       |      |
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|            |  |  |  | <p><b>1-3 marks</b><br/>Coverage of one section<br/><i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i></p> <p><b>0 marks</b><br/><i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i></p> |                 |     |     |       |       |      |
|            |  |  |  | Question 6 total                                                                                                                                                                                                                                                                                                                                                                                                                        | 4               | 5   |     | 9     |       |      |